

- 30** The transitions of electrons between three consecutive energy levels in an atom give rise to three spectral lines. The shortest and longest wavelengths of the spectral lines are λ_1 and λ_2 respectively.

Which of the following gives the wavelength of the other spectral line, in terms of λ_1 and λ_2 ?

A $\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)^{-1}$

B $\frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

C $\frac{\lambda_1 + \lambda_2}{2}$

D $\lambda_1 - \lambda_2$